


# AI-Powered Unified Sri Lankan Tourism Platform Using GraphRAG and LLM Orchestration

R26-IT-050

## Project Proposal Report

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Department of information technology
Sri Lanka Institute of Information Technology Sri Lanka

August 2026

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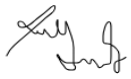
Department of information technology

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# Declaration

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The supervisors should certify the proposal report with the following declaration. The above Candidates are carrying out research for the undergraduate dissertation under my supervision.

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| Signature of Supervisor | | |
| Signature of Co-Supervisor | | |

# Abstract

Sri Lanka's tourism industry is rapidly expanding, yet the digital tools available to travelers remain fragmented and disconnected. Tourists frequently depend on multiple applications to arrange accommodation, transportation, and activities, often without real-time safety validation or integrated planning support. Most existing tourism systems focus on recommending individual services rather than generating complete and coherent travel packages that satisfy multiple user requirements simultaneously.

This research proposes the development of an intelligent tour package generation system based on a knowledge graph framework. Tourism entities such as hotels, drivers, vehicles, and activities are modeled as interconnected nodes, enabling structured reasoning across relationships such as location, availability, and service compatibility. User travel preferences—including destination, accommodation type, dietary requirements, and scheduling constraints—are transformed into structured conditions that guide graph traversal and constraint-based filtering.

To enhance reliability and reduce inconsistencies, the system employs a Graph-Based Retrieval-Augmented Generation (Graph RAG) approach, ensuring that Large Language Model (LLM) outputs are grounded in validated graph data. Additionally, real-time external APIs are integrated to assess traffic conditions, travel duration, and opening hours, ensuring that generated itineraries are practical and executable.

By combining knowledge representation, constraint satisfaction, grounded language generation, and real-time feasibility validations.

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# Introduction

Tourism is a vital contributor to Sri Lanka's economic growth, supporting employment, regional development, and foreign exchange earnings. As the industry continues to recover and expand, the number of international travelers visiting the country is steadily increasing. However, despite this growth, the digital tools available for travel planning remain fragmented. Tourists often depend on multiple separate applications to arrange accommodation, transportation, activities, and safety-related information. This lack of integration makes trip planning complex and time-consuming, especially for independent travelers seeking personalized experiences.

Most existing tourism platforms focus on recommending individual components such as hotels or attractions using basic filtering and ranking techniques. While these systems provide limited personalization, they do not generate fully integrated and coherent travel packages. Recommendations are often treated as independent suggestions rather than interconnected elements of a complete itinerary. In addition, many systems fail to validate practical constraints such as traffic conditions, travel duration, weather changes, and operating hours, which may result in unrealistic or impractical travel plans.

Recent advancements in Artificial Intelligence, particularly in Large Language Models (LLMs) and retrieval-based techniques, offer new opportunities for intelligent travel assistance. However, purely generative systems may produce inconsistent or unverified information if not grounded in structured data. Therefore, there is a need for a more reliable and context-aware solution that combines structured knowledge representation with intelligent reasoning and real-time validation.

This research proposes the development of a knowledge-graph-based intelligent tour package generator that produces customized and feasible travel itineraries. By integrating graph-based reasoning, constraint validation, grounded language generation, and real-time data from external sources, the system aims to deliver coherent, practical, and scalable tourism solutions tailored to Sri Lanka's evolving travel ecosystem.

## Background and Literature Survey

Tourism planning systems have evolved from simple listing platforms to personalized recommender systems that suggest hotels, attractions, or transport based on user preferences. Early research focused on optimization algorithms and constraint-based trip planners that arranged attractions efficiently, while more recent studies introduced Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) techniques to enhance personalization and conversational travel assistance. Additionally, Knowledge Graphs have been explored to represent structured relationships

among tourism entities, improving recommendation accuracy through semantic reasoning. However, existing systems largely treat tourism components as separate recommendations rather than generating fully integrated and constraint-validated tour packages. Many approaches also rely on static datasets and lack real-time feasibility validation such as traffic conditions, weather, or service availability. Furthermore, purely generative AI systems may produce unreliable outputs when not grounded in structured data. These limitations highlight the need for a graph-driven, constraint-aware, and real-time validated itinerary generation framework that combines knowledge representation with grounded LLM generation to deliver coherent, practical, and personalized tourism solutions.

Lu, Fang, and Tseng [1] proposed the Package-Attraction-based Trip Planner (PAT-Planner), which integrates tourist packages and attractions using constraint-based optimization. Their approach demonstrated the importance of satisfying travel constraints during itinerary generation. However, the system relied on scoring models and heuristic algorithms, lacking semantic reasoning capabilities across interconnected tourism entities. The use of static datasets also limited contextual adaptability and real-time validation.

With the advancement of Artificial Intelligence, Large Language Models (LLMs) have been increasingly applied to tourism systems. Recent research such as TravelRAG [2] integrates multi-layer knowledge graphs with Retrieval-Augmented Generation (RAG) to enhance tourism information retrieval. By leveraging structured graph traversal before generative output, TravelRAG improves factual consistency and reduces hallucinated responses. Similarly, TP-RAG [3] benchmarks retrieval-augmented LLM frameworks for spatiotemporal-aware travel planning, highlighting the

importance of grounding LLM outputs in structured data for reliable itinerary reasoning.

Other studies, including Att-KGCN [4], explore knowledge graph-based recommendation using graph convolution networks to improve attraction prediction accuracy. While these models demonstrate the effectiveness of knowledge graphs in capturing relational dependencies, they focus primarily on recommendation ranking rather than generating complete, constraint-validated tour packages. Additionally, TourLLM [5] illustrates the potential of domain-specific fine-tuning for tourism knowledge integration but does not incorporate explicit graph-based constraint satisfaction mechanisms.

Despite these advancements, existing systems exhibit several limitations. Many LLM-based solutions rely on unstructured retrieval pipelines, which may produce coherent yet impractical itineraries when real-time constraints such as travel time, traffic conditions, and service availability are not validated. Furthermore, most approaches generate recommendations for individual components rather than ensuring logical coherence across accommodation, transport, and activity selections as a unified package.

Therefore, while prior research demonstrates the value of knowledge graphs and RAG frameworks in tourism applications, there remains a research gap in developing a GraphRAG-driven, constraint-satisfaction-based itinerary generation system that integrates multi-criteria reasoning with real-time feasibility validation. This study aims to address this gap by combining structured knowledge graph traversal, grounded LLM generation, and dynamic API-based validation to produce coherent, executable, and personalized tour packages.

## Research Gap

Existing literature demonstrates significant progress in both constraint-based trip planning and knowledge-enhanced AI systems for tourism applications. Lu et al. [1] proposed a constraint-aware trip planner that integrates tourist packages and attractions; however, their approach relies on heuristic scoring and optimization methods without leveraging semantic knowledge graph reasoning. While effective in satisfying predefined travel constraints, the model does not incorporate relational multi-hop reasoning across interconnected tourism entities.

Recent research has explored the integration of knowledge graphs and Retrieval-Augmented Generation (RAG) to enhance information retrieval accuracy. TravelRAG [2] demonstrates how multi-layer knowledge graphs improve factual consistency in tourism-related queries, while TP-RAG [3] introduces spatiotemporal-aware retrieval frameworks for travel planning tasks. These studies highlight the importance of structured retrieval before generative output. However, their focus remains primarily on information retrieval and recommendation accuracy rather than full itinerary orchestration with strict constraint validation.

Similarly, graph-based recommendation models such as Att-KGCN [4] show the effectiveness of knowledge graphs in improving attraction recommendation performance through graph convolution techniques. TourLLM [5] further demonstrates the benefit of tourism-specific LLM enhancement. Nevertheless, these approaches emphasize ranking and personalization rather than generating fully integrated, multi-criteria tour

packages where accommodation, transport, activities, and scheduling must function coherently as a single executable plan.

Moreover, although RAG-based LLM systems improve conversational interaction, most existing implementations rely on document-based retrieval pipelines rather than structured graph traversal combined with constraint-satisfaction mechanisms. Real-time feasibility validation—such as verifying travel time, traffic conditions, and operational hours—remains largely unaddressed in current tourism AI research.

Therefore, despite advancements in constraint-based planning, knowledge graph modeling, and RAG-enhanced generative systems, there remains a gap in developing a unified GraphRAG-driven framework that integrates semantic graph reasoning, multi-criteria constraint validation, grounded LLM generation, and real-time feasibility checking to produce coherent and practically executable tour packages. This research seeks to bridge this gap by proposing a structured, graph-based itinerary generation model tailored to dynamic tourism environments.

## Research Problem

Despite advancements in tourism recommender systems and AI-driven travel assistance, existing digital platforms remain limited in their ability to generate fully integrated and practically feasible tour packages. Most current systems focus on recommending individual components—such as hotels, attractions, or transport—using filtering, ranking, or scoring mechanisms. While these approaches support basic personalization, they do not ensure that selected components function coherently as a unified itinerary.

Traditional trip planning models rely primarily on optimization algorithms and predefined constraints, but they lack structured semantic reasoning across interconnected tourism entities. More recent AI-based systems incorporating Large Language Models (LLMs) enhance conversational interaction; however, purely generative approaches may produce inconsistent or unverified outputs when not grounded in structured data. Furthermore, many existing solutions fail to validate real-time contextual constraints such as traffic conditions, travel duration, service availability, and operational hours before presenting recommendations to users.

As a result, travelers may receive suggestions that satisfy individual preferences but are impractical, incomplete, or difficult to execute in real-world scenarios. Therefore, there is a need for an intelligent, graph-driven, and constraint-aware framework capable of integrating structured tourism data, multi-criteria reasoning, grounded generative outputs, and real-time feasibility validation to produce coherent, reliable, and executable tour packages.

# Objectives

## Main objective

The main objective of this research is to design and develop an intelligent, knowledge-graph-driven tour package generation framework that utilizes Graph-Based Retrieval-Augmented Generation (Graph RAG) to produce personalized, multi-criteria, and practically feasible travel itineraries.

The proposed system aims to integrate structured tourism data—including hotels, transport services, activities, and related entities—within a knowledge graph, enabling semantic reasoning across interconnected components. It further seeks to translate user travel preferences into structured constraints, perform graph-based traversal and constraint validation, and generate coherent tour packages grounded in verified data.

Additionally, the framework intends to incorporate real-time contextual information such as traffic conditions, travel duration, and service availability to ensure that generated itineraries are not only personalized but also operationally executable in real-world scenarios. Through this approach, the research aims to address the limitations of traditional recommended systems by producing integrated, reliable, and constraint-aware tour packages rather than isolated travel recommendations.

## Specific objectives

1. **To collect and define structured tourism data** including hotels, drivers, vehicles, and activities, and identify key attributes such as location, availability, service type, pricing category, and operational constraints required for itinerary generation.

2. **To design and implement a domain-specific Knowledge Graph** that represents tourism entities as nodes and their relationships (e.g., location, availability, proximity, service compatibility) as edges to enable structured semantic reasoning.

3. **To develop a mechanism for translating user travel preferences into structured constraints**, converting natural language inputs such as destination, accommodation type, dietary preferences, activity interests, and transport requirements into machine-readable conditions.

4. **To implement graph-based traversal and constraint-satisfaction algorithms** capable of identifying compatible combinations of hotels, activities, transport services, and other tourism components that collectively satisfy all specified user requirements.

5. **To integrate Graph-Based Retrieval-Augmented Generation (Graph Rag)** in order to ground Large Language Model (LLM) outputs in validated graph-selected data, thereby improving reliability and minimizing hallucinated or inconsistent itinerary generation.

6. **To incorporate real-time external data sources**, such as traffic conditions, travel time, and service operating hours, to validate the practical feasibility and executability of generated itineraries.

7. **To evaluate the effectiveness of the proposed framework** in generating coherent, personalized, and operationally feasible tour packages compared to traditional recommendation-based approaches.

## Methodology

This research adopts a design and development–oriented methodology to construct and evaluate a knowledge-graph-driven intelligent tour package generation system. The methodology consists of five main **phases**: data modeling, knowledge graph construction, constraint processing, GraphRAG-based itinerary generation, and real-time feasibility validation.

### Data Collection and Attribute Definition

The initial phase involves collecting representative tourism data, including hotels, drivers, vehicles, attractions, and activities. Each entity will be defined with structured attributes such as location, service type, availability, pricing category, operational hours, dietary options, and certification status. This structured data will serve as the foundation for itinerary reasoning and package generation. The focus will be on designing a schema that supports

multi-criteria filtering and constraint validation rather than relying on large pre-trained datasets.

## Knowledge Graph Design and Implementation

A domain-specific Knowledge Graph will be developed to represent tourism entities and their relationships.

- **Nodes** will represent entities such as hotels, transport services, activities, and locations.
- **Edges** will represent relationships such as located in, offers service, available on, near to, and operates during.

The graph structure enables multi-hop reasoning, allowing the system to identify compatible combinations of tourism components based on relational dependencies. This structured representation supports semantic traversal rather than simple keyword matching.

## User Preference Structuring and Constraint Translation

User inputs will be collected through a structured interface where travelers specify preferences such as destination, accommodation type, food requirements, activity interests, transport needs, and travel dates.

These inputs will be transformed into machine-readable constraints. For example, natural language requirements will be mapped into structured query conditions that can be evaluated against the knowledge graph. This step ensures compatibility between user preferences and graph-based reasoning mechanisms.

## GraphRAG Integration and LLM-Based Generation

The proposed framework incorporates Graph-Based Retrieval-Augmented Generation (GraphRAG). Instead of allowing unrestricted generative responses, the Large Language Model (LLM) will operate strictly on graph-validated results.

The knowledge graph will serve as the retrieval layer, while the LLM will generate structured, human-readable itinerary descriptions based on selected data. This grounding mechanism improves reliability and reduces hallucinations.

## System Architecture Overview

The proposed system consists of four main layers:

1. **Frontend Layer (Web & Mobile)**
    - Web application developed using **React.js**
    - Mobile application developed using **React Native**
    - Collects user preferences and displays generated itineraries
2. **Backend Layer**
    - Developed using **Node.js and Express.js**
    - Handles API requests, business logic, constraint processing, and GraphRAG integration
3. **Database and Knowledge Graph Layer**
    - **MongoDB** used for storing structured tourism data
    - Knowledge graph constructed using graph-based data structures within the backend logic

- Stores relationships between hotels, drivers, vehicles, activities, and locations

4. **AI & External Integration Layer**
    - GraphRAG framework for structured retrieval
    - LLM API for grounded itinerary generation
    - External APIs for traffic, travel time, and operational hour validation

# Diagrams and Charts

## System architecture diagram

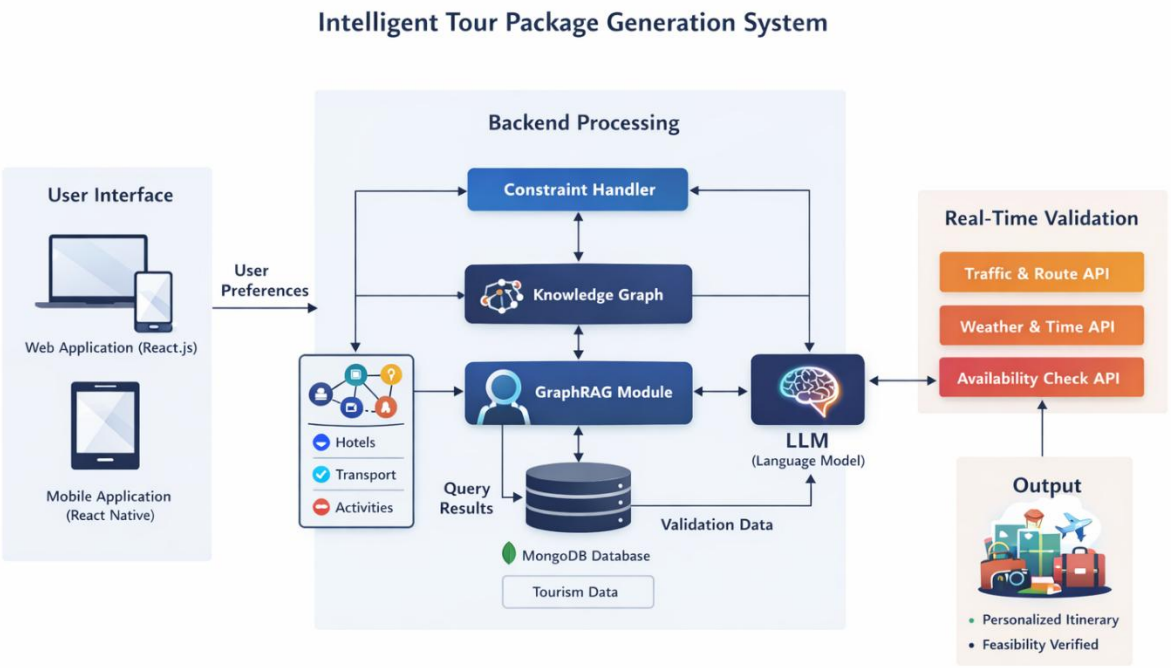



*Figure 1 System Architecture Diagram*

## Component overview diagram

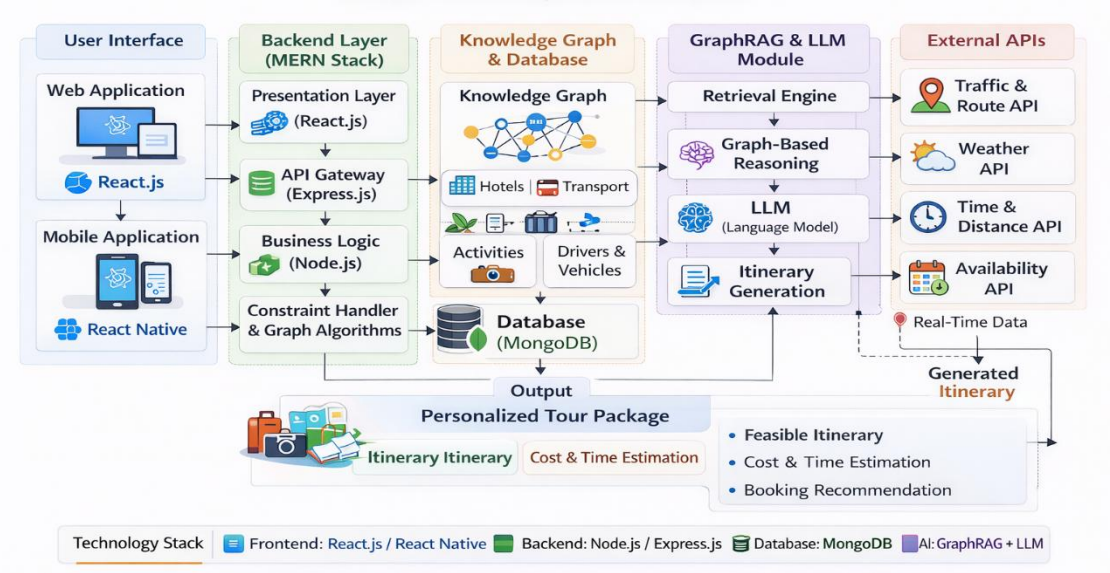



*Figure 2 Component overview diagram*

## Work breakdown structure

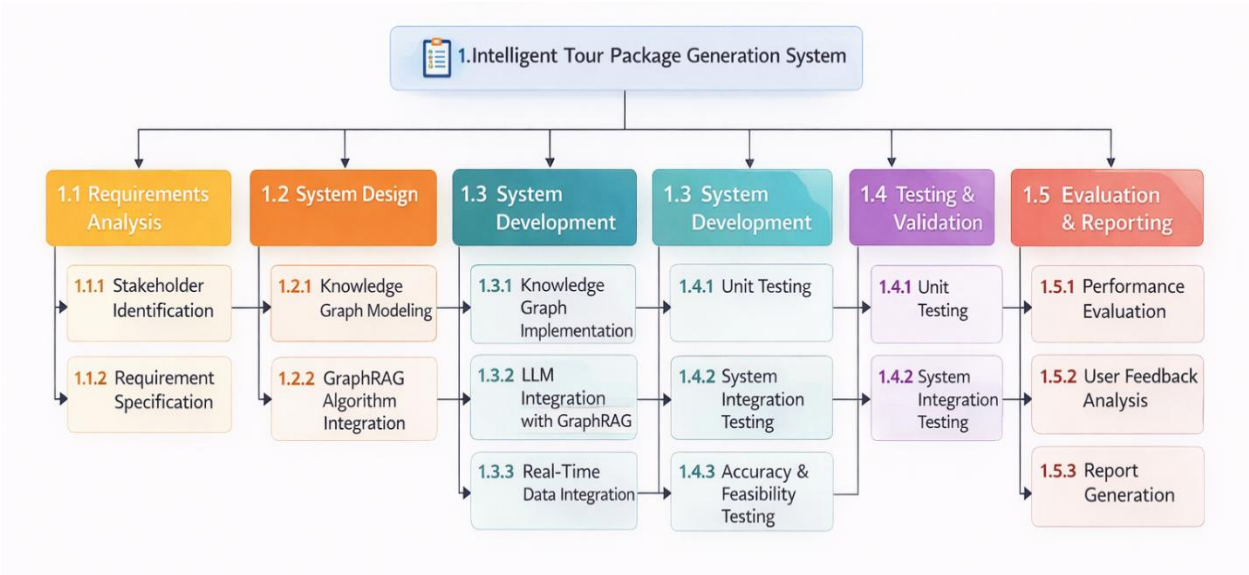



*Figure 3 Work breakdown structure*

## Grantt Chart

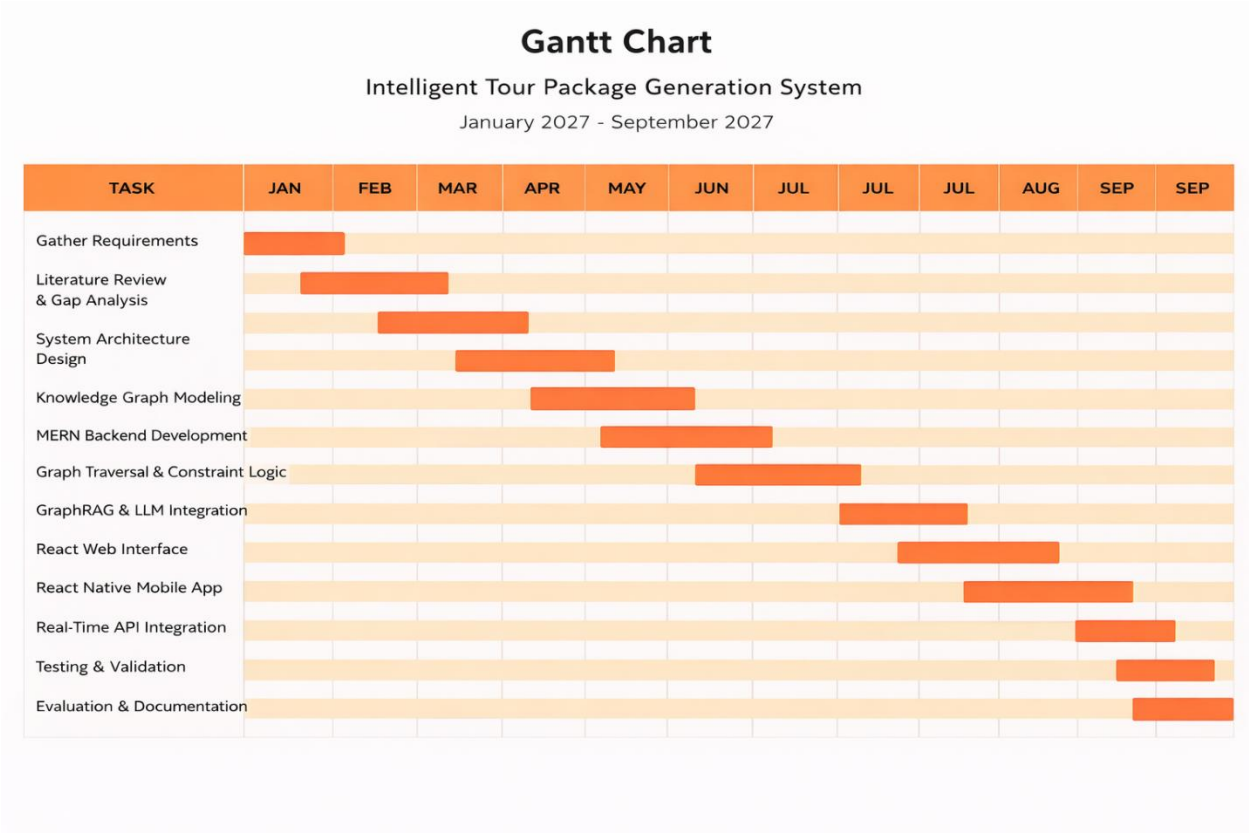



*Figure 4 Grantt Chart*

# Project Requirements

## Functional requirements

- **User Input Processing**

1. The system shall allow users to input travel preferences including destination, travel dates, accommodation type, transport preferences, activities, and budget constraints.
2. The system shall parse and convert user inputs into structured, machine-readable constraints.
3. The system shall validate mandatory inputs (e.g., destination, dates) before processing itinerary generation.

- **Tourism Data Management**

1. The system shall store structured tourism data including hotels, drivers, vehicles, and activities.
2. The system shall define and maintain attributes such as location, availability, service type, operating hours, and pricing category.
3. The system shall represent tourism entities and relationships within a knowledge graph structure.

- **Knowledge Graph Reasoning**

1. The system shall perform graph traversal to identify compatible tourism components based on user-defined constraints.

2. The system shall ensure that selected hotels, transport services, and activities are logically connected (e.g., location compatibility).
3. The system shall support multi-hop reasoning across entities to satisfy multiple criteria simultaneously.

- **Constraint-Satisfaction Validation**

1. The system shall validate that all user-specified constraints are satisfied before generating the final itinerary.
2. The system shall eliminate combinations that violate availability, schedule, or compatibility constraints.
3. The system shall prevent generation of incomplete or partially matched tour packages.

- **GraphRAG & LLM Integration**

1. The system shall retrieve structured results from the knowledge graph as input to the GraphRAG module.
2. The system shall ensure that the LLM generates itinerary descriptions strictly based on graph-validated data.
3. The system shall produce human-readable, coherent, day-wise itinerary descriptions.

- **Real-Time Feasibility Validation**

1. The system shall integrate external APIs to obtain real-time traffic and route information.
2. The system shall validate travel time between selected locations.

3. The system shall check availability and operating hours of selected services.
4. The system shall update or reject itineraries that are not practically feasible.

- **Output Generation**

1. The system shall generate a complete, integrated tour package including accommodation, transport, and activities.
2. The system shall present a day-wise schedule with estimated travel time and feasibility confirmation.
3. The system shall provide cost and time estimation for the itinerary generated.

## Non-Functional Requirements

- **Performance Requirements**

1. The system shall generate a personalized itinerary within an acceptable response time (e.g., ≤ 5–8 seconds under normal network conditions).
2. Graph traversal and constraint validation shall be executed efficiently to prevent excessive computation delays.
3. Real-time API calls shall be optimized to minimize latency during feasibility validation.

- **Reliability Requirements**

1. The system shall ensure that LLM-generated outputs are grounded in graph-validated data to minimize hallucinations.
2. The system shall prevent incomplete itinerary generation when required data is unavailable.
3. The system shall log in errors and failed API responses for monitoring and debugging.

- **Usability Requirements**

1. The web and mobile interfaces provide a clear and intuitive user experience for entering travel preferences.
2. Generated itineraries shall be presented in a readable, structured, and user-friendly format.
3. The system shall provide meaningful feedback if user constraints cannot be satisfied.

- **Accuracy & Quality Requirements**

1. The system should ensure high constraint satisfaction accuracy when matching user preferences.
2. Real-time validation results reflect up-to-date information from integrated external APIs.
3. The generated itinerary shall maintain logical coherence across selected tourism components.

## System Requirements

1. **Software Requirements**

- Frontend: React.js (Web Application) and React Native (Mobile Application)
- Backend: Node.js with Express.js
- Database: MongoDB for structured tourism data storage
- AI Integration: Large Language Model (LLM) API and GraphRAG implementation
- External APIs: Traffic, weather, time estimation, and availability validation services
- Development Tools: Visual Studio Code or equivalent IDE

2. **Hardware Requirements**

- Development machine with minimum 8GB RAM and modern multi-core processor
- Internet connectivity for API and LLM access
- Cloud or server environment for deployment and scalability

# Description of Personal and Facilities

| Registration Number | Name | Task description |
|---|---|---|
| IT 22895196 | Deniyawatte D G A D N | • Define and structure tourism-related entities such as hotels, drivers, vehicles, and activities, and identify key attributes (e.g., location, availability, service type, operating hours) required for multi-criteria itinerary generation.<br><br>• Construct a domain-specific knowledge graph representing tourism entities as nodes and their relationships as edges, enabling semantic connections and multi-hop reasoning across interconnected components.<br><br>• Convert user travel preferences into structured constraints and perform graph traversal and filtering to identify compatible |

| | | |
|---|---|---|
| | | combinations of accommodation, transport, and activities that satisfy all specified criteria simultaneously.<br><br>• Apply Graph-Based Retrieval-Augmented Generation (GraphRAG) to retrieve validated graph results and generate coherent, human-readable itineraries using a Large Language Model while minimizing hallucinated outputs.<br><br>• Integrate external APIs to validate traffic conditions, travel time, availability, and operational hours, ensuring that generated itineraries are practically executable in real-world scenarios. |

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